Lecture 9 and 10a

- more EU applications
- quantitying risk
- quantifying risk aversion
- paradoxes

Another EU Application: Value of Life

Consider a risk-averse individual who is offered these two choices:

- safe option: he goes through his day as normal, and his current wealth is w_0
- risky option: he accepts a payment of $\$\Delta$ to do a task, which is costless, but will kill him with a very small probability p
- If he takes the risky option, find an upper bound on the financial value L he places on his life
  - assume that dying is like losing $\$L$, and that (questionable, yes :)) utility of wealth is the same whether alive or dead

Formal Solution

- An expected-utility maximizer chooses the option with the best EU
- If he takes the risky option,
  - with chance p he's killed, for $u(w_0 + \Delta - L)$
  - with chance $1 - p$, he gets $u(w_0 + \Delta)$
  - so $EU = pu\left(w_0 + \Delta - L\right) + (1 - p)u(w_0 + \Delta)$
- if he takes the safe option, he gets $u\left(w_0\right)$

Solution Cont'd

- since we're told he takes the risky option, it must be

$$\begin{aligned}
u(w_0) &\leq pu(w_0+\Delta-L)+(1-p)u(w_0+\Delta) \\
&< u\left(p(w_0+\Delta-L)+(1-p)(w_0+\Delta)\right) \text{ by risk-aversion} \\
&= u\left(w_0+\Delta-pL\right) \\
&\Rightarrow w_0 \leq w_0+\Delta-pL \\
&\Rightarrow L < \Delta/p
\end{aligned}$$

- intuitive solution:: he expects to gain wealth $\Delta - pL$ with the risky option
- but the risk involved is bad, so a risk-averse person must expect to strictly gain money, ie $\Delta - pL > 0$

Risk and Related Concepts

What Is Risk?

- we noted last week that \$200 for sure seems less risky than ending up with \$300 w.p. $\frac{1}{2}$, and \$100 w.p. $\frac{1}{2}$
- intuitively, the more spread out are your possible outcomes, the more risk you're facing
- mathematically, one way to capture this is by calculating the *variance*, or its square root, *standard deviation*

Variance and Standard Deviation

- consider a random variable X which takes on possible values $x_1, x_2, ..., x_n$, with probabilities $p_1, p_2, ..., p_n$
- the *variance:*

$$\begin{aligned} Var(X) &= E[X^2] - (E[X])^2 \\ &= E[(X - E[X])^2] \end{aligned}$$

- the *standard deviation:* $\sqrt{\text{variance}}$

Example

- if X has only one possible value, \$200, which happens with probability 1, then

$$E[X] = \$200, \;\; Var(X) = 0$$

- if X takes on two possible values, either \$100 or \$300 equally likely, then $E[X] \equiv \overline{X} = 200$
- variance of X with 1st formula:

$$E[X^2] - \overline{X}^2 = \frac{1}{2}(100)^2 + \frac{1}{2}(300)^2 - 200^2 = 10\,000$$

- variance of X with 2nd formula:

$$E[(X - \overline{X})^2] = \frac{1}{2}(300 - 200)^2 + \frac{1}{2}(100 - 200)^2 = 10\,000$$

Next Definition: Certainty Equivalent

Definition

Given a risky gamble that yields gains $X_1, \ldots, X_n$ with probabilities $p_1, \ldots, p_n$, and an initial wealth w_0, the certainty equivalent (CE) is the lump-sum amount that's exactly as good as the gamble:

$$u(w_0 + CE) = \underbrace{p_1 u(w_0 + X_1) + p_2 u(w_0 + X_2) + \ldots + p_n u(w_0 + X_n)}_{EU \text{ of risky gamble}}$$

Definition

The risk premium for the above gamble is $\pi = E[\text{wealth gain with gamble}] - CE$

- In other words, $w_0 + CE \sim$ gamble, and the risk premium is the amount of wealth the DM would sacrifice to eliminate risk:

  $u(w_0 + \text{exp wealth gain of gamble } - \pi) = EU \text{ of gamble} = u(w_0 + CE)$

Example

- Melsi has initial wealth \$200, and plays a gamble with a half chance of losing \$100 and a half chance of winning \$200
- if $u(w) = \sqrt{w}$, her CE solves

$$\begin{aligned} \sqrt{200 + CE} &= \frac{1}{2}\sqrt{200 - 100} + \frac{1}{2}\sqrt{200 + 200} \\ \Rightarrow & \quad 200 + CE = 15^2, \text{ so } CE = \$25 \end{aligned}$$

- the expected wealth gain from the gamble is $\frac{1}{2}(-100) + \frac{1}{2}(200) = 50$
- the difference is her risk premium, $\pi = 50 - 25$
- interpretation: Melsi would take a flat payment of \$25 in lieu of the gamble, where she expects to gain \$50; she's evidently willing to sacrifice the difference, \$25, to eliminate the risk

Observations

- a risk-averse person must be compensated to take on risk, thus their risk premium is positive and their CE is smaller than the expected wealth gain:

$$u(w_0 + CE) \equiv \underbrace{EU \text{ of gamble } < u(w_0 + \text{exp wealth gain})}_{\text{by risk aversion}}$$

  - comparing the LHS and RHS expressions, $CE <$ exp wealth gain, so $\pi =$exp wealth gain $- CE > 0$
- for a risk-lover, CE exceeds the gamble's expected wealth gain, so the risk premium is negative: instead of sacrificing wealth to remove the risk, you'd need a bonus to compensate for taking the fun away!
- for a risk-neutral person, CE $=$ expected wealth

Quantifying Risk Aversion

- two measures of risk-aversion:
  - coefficient of absolute risk aversion: $\rho_A^u(w) = -u''(w)/u'(w)$
  - coefficient of relative risk aversion: $\rho_R^u(w) = -wu''(w)/u'(w)$
- some important utility functions in applications:
  - CARA (constant absolute risk aversion: $u(w) = 1 - e^{-\alpha w}$
    - here, $u' = \alpha e^{-\alpha w}$, $u'' = -\alpha^2 e^{-\alpha w}$, so $-u''/u' = \alpha$
  - CRRA (constant relative risk aversion: $u(w) = w^{1-\alpha}/(1-\alpha)$ if coefficient of relative risk aversion is $a \neq 1$, and $u(w) = \ln w$ has a coefficient of 1

Arrow-Pratt Theorem

The following are equivalent ways to say that DM #1 is more risk-averse than DM #2:

1. DM #1 has a higher coefficient of absolute risk aversion at *all* wealth levels w
2. DM #1 has a higher risk premium for *every* gamble
3. DM #1's utility-of-wealth function is more concave than DM #2's

Risky Assets

- a *risky asset* is an asset that has various possible returns, depending on the "state of the world"
- for example, in a gamble where you could either win money or lose money, there is a "win" state and a "loss" state
- to be concrete, assume there are:
  - possible states $s = 1, 2, ..., S$
  - the probability of state s is p_s
  - the *return rate* in state s is r_s

Return Rates and Risk

- a *return rate* r_s means that if you invest \$m, you gain (or lose) $r_s m$
- so, at the end of the day, your \$m is worth $m + r_s m = (1 + r_s)m$
- for example, if you invest \$100 and get a return rate of 0.1, or 10%, it means you gain $.1(100) = \$10$, and end up with \$110
- and recall that we measure risk by $\sigma^2 = \sum_s p_s(r_s - \bar{r})^2 = \sum p_s r_s^2 - \bar{r}^2$, where $\bar{r} = \sum_s p_s r_s$ is the expected return rate; risk tells you how spread out the returns are, compared to the average

Return Rates and Risk

- most people dislike risk (they are "risk-averse"), and so, for two investments with the same expected return rate, they will choose the one with less risk (lower σ)
- most people like money, and so, for two investments with the same risk, they will choose the one with the higher expected return rate
- usually, there is a trade-off:
    - higher mean return rate is good
    - higher variance is bad
    - most people try to optimally balance the two, given a constraint that to get a higher return, you usually have to accept higher risk

Example

Suppose that there are two assets available: a risk-free asset, with return rate $r^f = .05$, and a risky asset with expected return $r^M = 0.2$, risk (standard deviation) $\sigma^M = 0.12$. If a consumer wants to invest \$100 and earn an expected wealth increase of \$15, what risk σ must he accept on his overall investment?

Useful Facts:

If you invest a fraction α of your wealth in the safe asset, and a fraction $1 - \alpha$ of your wealth in the risky asset, then:

- expected return rate is just the weighted average $\alpha r^f + (1 - \alpha) r^M$
- overall risk (standard deviation) is $(1 - \alpha)\sigma^M$
- you may just take my word for this :), though it's fairly easy to show
- a bit more complicated with multiple risky assets, your overall risk then depends also on how correlated they are

Solution

- An expected wealth increase of \$15 on a \$100 investment means a return rate of 15%
- so we need the fraction $(1-\alpha)$ in the risky asset to solve $\alpha(.05)+(1-\alpha)(.2)=.15 \Rightarrow \alpha=\frac{1}{3}$
- thus, he invests $1-\alpha-\frac{2}{3}$ of his wealth in the risky asset, for overall risk (using useful facts above) $\sigma=\frac{2}{3}(.12)=.08$

Paradoxes

- we might not get to all of these
- leaving them for interest, and may also discuss some in the midterm review lecture

Ellsberg Paradox

- suppose there is an urn with 300 balls
- you are told that 100 of the balls are red, while each of the remaining 200 balls is blue or yellow (unknown how many are blue and how many are yellow)
- given a choice between the following two lotteries, most people strictly prefer the first one:
  - p^1: one ball will be drawn. You get a prize if it's red, but \$0 otherwise
  - p^2: one ball will be drawn. You get a prize if it's blue, but \$0 otherwise
- that is, people prefer betting on red to blue

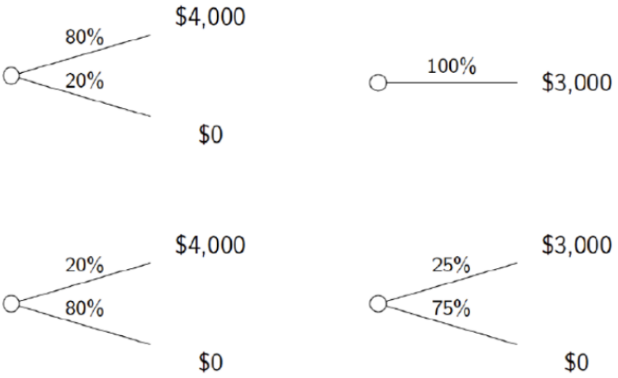
Ellsberg Paradox, Cont'd

- however, most people also strictly prefer betting on $(Y \cup B)$, to betting on $(Y \cup R)$
- but these two preferences are inconsistent with our theory
- for suppose your beliefs put chance p_B on a randomly drawn ball being blue, and p_Y on yellow (it is given that $p_R = \frac{100}{300} = \frac{1}{3}$, so we know $p_B + p_Y = \frac{2}{3}$)
- to bet on red over blue, it must be that $p_R > p_B$, i.e. $p_B < \frac{1}{3}$
- but to bet on $(Y \cup B)$ over $(Y \cup R)$, it must be that $p_Y + p_B > p_Y + \frac{1}{3} \Leftrightarrow p_B > \frac{1}{3}$
- opposite!

Presumed Explanation for Ellsberg

- most people are *ambiguity-averse*
- betting on red is more appealing than betting on blue, because there is no ambiguity about your chances of winning on red: it is $\frac{1}{3}$
- and similarly, betting on yellow or blue is appealing, because the probability of a win is again objectively given as $\frac{2}{3}$
- there is a large literature on ambiguity aversion, including a max-min representation theorem: an ambiguity-averse person can be described as choosing the worst-case-scenario beliefs in each problem, and choosing accordingly
- for example, worst-case scenario in betting on $Y \cup R$ is that there are no yellow balls in the urn; while worst-case scenario in betting on B is that there are ONLY yellow balls in the urn

Allais Paradox: Certainty Effect



Allaix Paradox

- recall: most people prefer the top right gamble, but the bottom left one
- these preferences are inconsistent with expected utility maximization, and in particular violate the independence axiom
- what might people actually be doing?

Proposal #1: Certainty Effect

- a simple explanation: people are basically EU-maximizers, but treat *certain* outcomes differently
- so they pick \$3000 for sure because it gets a bonus for having no chance of loss

Related Puzzle

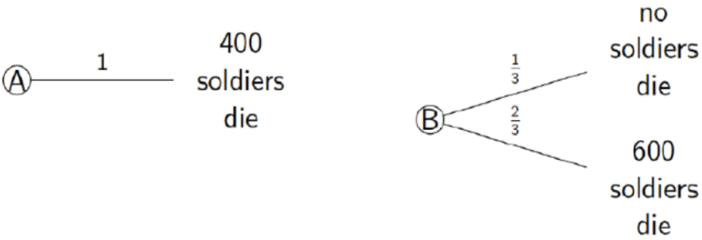
- we made an innocuous-sounding assumption, “consequentialism”, which said that the way a lottery is described shouldn’t matter
- but in fact, this is often an issue!
- here’s an illustration with the Allais paradox

Isolation Effect

- Consider the following 2-stage game:
  - stage 1: 75% chance of ending with nothing, 25% chance of moving onto stage 2
  - stage 2: choice of \$3000 for sure, or $\begin{cases} 4000 & \text{w.p. } 0.8 \\ 0 & \text{w.p. } 0.2 \end{cases}$
- Most choose the option with \$3000 for sure in stage 2
- But in terms of final outcomes, they are NOW choosing a .25 chance at \$3000 over a $.25(.8) = .2$ chance at \$4000; a flip of the initial preferences!

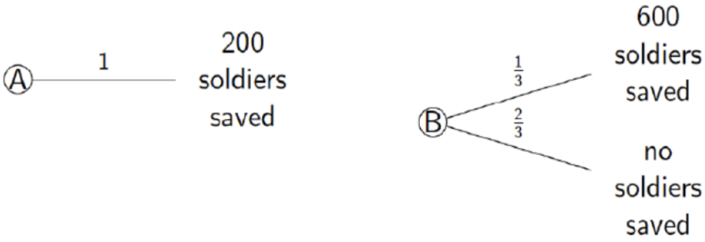
Framing

You are an army general, and can take 2 routes:



Framing Cont'd

Compare to: you are a general in an army, and 600 soldiers have been ambushed. Two options:



Framing Cont'd

- second problem, most people pick saving 200 over the risky choice
- first problem, most people pick risky choice over killing 400 for sure
- but they are the **same!**
  - A in the first problem (kill 400 of 600) = A in the 2nd problem (save 200 of 600)
  - B in the first problem ($\frac{2}{3}$ chance of kill everyone, $\frac{1}{3}$ chance of kill nobody) = B in 2nd problem ($\frac{2}{3}$ chance of save nobody, $\frac{1}{3}$ chance of save everybody)

Framing Cont'd

- lesson: the framing of the problem can affect preferences! (in this case, by influencing which outcomes are perceived as gains, and which are perceived as losses)
- note: this problem also shows that people tend to be risk-loving (prefer the gamble) over bad stuff, but risk-averse (prefer the sure thing) over good stuff

Intro to Prospect Theory

This is an alternative model proposed by Kahneman (1979), which is similar to expected utility theory, but:

- focuses on gains and losses (relative to some reference point), rather than on final outcomes
- postulates a valuation function for gains and losses (as opposed to a utility-of-wealth function), which is concave for gains, convex for losses
- allows probabilities to be weighted
- allows for an "editing" phase, where framing can influence how people select a reference point and simplify the gamble

Prospect Theory: Editing

When presented with a gamble, the first step is "editing":

- choose a **reference point**, so as to identify the gains vs losses
- simplify: combine outcomes with the same payoff (e.g. if you will win \$20 if a dice comes up on either a 1 or a 5, view this as winning \$20 w.p. $\frac{2}{6}$)
- simplify ("segregation/cancelation"): cancel out the parts that are the same, and focus only on differences
- look for dominance: choose a gamble if it clearly dominates another

Prospect Theory: After Editing

The basic ingredients to evaluate a gamble are:

- a "probability weighting" function, which transforms a probability p into a weighted probability $\pi(p)$
  - assume: $\pi(0) = 0$, $\pi(1) = 1$
  - typically assume: small probabilities are overweighted, large probabilities are underweighted
- a function v to evaluate gains and losses, relative to the reference point
- if there are 2 outcomes, a gain x (chance p) and a loss y (chance q), prospect theory says your valuation is

$$\pi(p)v(x) + \pi(q)v(y)$$